

DriftTrace IDS2018 Reproduction

Complete record of all experiments, configurations, results and findings

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August 2026

1. Scope and Objective

This report documents a systematic attempt to independently reproduce the IDS2018 experimental results from: Pan et al., "DriftTrace: Combating Concept Drift in Security Applications Through Detection and Explanation", IEEE TIFS vol. 21, 2026. All 19 experiment versions are documented here, from the initial baseline (V1, global temporal split, squared distance) through to the final combined detection-and-explanation scripts (V11, V12). Each version isolates one variable change to identify its contribution.

2. Fixed Configuration (All Versions)

The following components remain identical across all versions:

Component	Value	Source
Dataset	CSE-CIC-IDS2018, 3 files, 147,199 rows, 4 classes	Paper Table IV
Architecture	83→64→32→16→3 encoder+decoder	Paper Section III-B1
Activation	ReLU on hidden layers, linear on bottleneck and output	Paper explicit
Optimizer	Adam	Paper explicit
Lambda λ	0.1	Paper Eq. 3
Batch size	256 for IDS2018	Paper Section IV-A
IQR coefficient p	1.5	Paper Algorithm 1
Feature engineering	76 numerical + Timestamp_ms + DstPort OHE + Proto OHE	CADE Appendix 1
Scaler	MinMaxScaler [0,1] on training rows only	CADE Appendix 1
Algorithm 1	Exact paper pseudocode	Paper Section III-B2
Algorithm 2	Exact paper pseudocode (V11 onward)	Paper Section III-C

3. Experiment Versions — Detailed Record

Version V1 — Baseline — Global temporal split, squared distance d^2 , all-pairs sum

Files: phase2_v1.py + phase3_v1.py / phase3_v2.py

Phase 2: Global sort all 147,199 rows by Timestamp, cut at row 117,759 (80%). Problem: DoS-Hulk (all Feb 16) and Infiltration (all Mar 1) land entirely in training — hold-out scenarios for these classes have zero drift test rows.

Phase 3: Squared distance d^2 in contrastive loss. All-pairs sum normalisation. BALANCED=True. 80/20 hold-out (contaminated — drift class partially in training).

Metric	Value
Benign F1	0.655
Bot F1	0.554
DoS-Hulk F1	0.996
Infiltration F1	0.206
Average F1	0.603
Δ paper	-0.367

Finding: Global temporal split makes DoS-Hulk and Infiltration hold-out scenarios trivially easy (all test samples are from the same class) but only because the model trained on them. The high DoS-Hulk F1=0.996 is misleading — DoS-Hulk was in training. Fundamental protocol error identified.

Version V2/V3 — V2: Same as V1. V3: BALANCED=False test

Files: phase2_v2-v6.py + phase3_v2.py / phase3_v3.py

Phase 2: Same global temporal split as V1 across V2-V6.

Phase 3: V2: same as V1. V3: BALANCED=False — tests whether paper's 'class-balanced' statement is critical.

Metric	Value
V2 Average F1	0.603
V3 Average F1	0.573
V3 Infiltration F1	0.118 (worst)
Δ paper	-0.397 to -0.367

Finding: BALANCED=False significantly worse. Without equal class representation, Infiltration (8% of data) gets ~20 batch slots — insufficient contrastive pairs. Paper's class-balanced mini-batch is essential.

Version V4 — L2 normalisation on encoder output — geometric incompatibility

Files: phase2_v4.py + phase3_v4.py

Phase 2: Same global temporal split.

Phase 3: Added $F.\text{normalize}(z, p=2, \text{dim}=1)$ to encoder output. Hypothesis: L2 norm projects all latent vectors onto unit sphere, preventing representation collapse.

Metric	Value
All 4 runs	STUCK at $\text{contras} \approx 4.67$
Average F1	~ 0.45
Δ paper	-0.52

Finding: L2 normalisation is geometrically incompatible with margin $m=10$. Unit sphere max squared distance = $4 < \text{margin}=10$. Contrastive loss cannot converge — different-class pairs can never reach $\text{sq_d} \geq 10$.

Version V5/V6 — V5: `similar_ratio=0.25` WRONG. V6: `similar_ratio=0.25` CORRECT

Files: `phase2_v5-v6.py` + `phase3_v5.py` / `phase3_v6.py`

Phase 2: Same global temporal split.

Phase 3: V5: attempted to limit negative pairs to achieve 25% positive ratio. Bug: $n_neg_target = n_pos \times 3 = 32,130 > \text{actual } n_neg = 21,675 \rightarrow$ used all negatives $\rightarrow$ ratio unchanged at 33% = identical to all-pairs. V6: correct implementation — subsample POSITIVE pairs. $n_pos_used = n_neg \times (0.25/0.75) = 7,225$ from 10,710 available.

Metric	Value
V5 Average F1	0.531 — identical to all-pairs
V6 Average F1	0.531 — IQR slightly wider but no F1 improvement
Key metric	IQR_Bot: $0.0003 \rightarrow 0.0014$, upper: $0.016 \rightarrow 0.025$

Finding: Correct ratio implementation confirmed. However temporal split still causes train/test distribution gap — last 20% of each class has different statistics from first 80%, pushing test samples outside IQR bounds computed from training.

Version V7 — Random stratified split — explored to fix precision collapse

Files: `phase2_v7.py` + `phase3_v7.py`

Phase 2: `sklearn train_test_split` with `stratify=class`, `random_state=42`. Each class contributes exactly 20% to test regardless of timestamp.

Phase 3: Same as V6 otherwise.

Metric	Value
Average F1	0.568
Benign precision	$0.489 \rightarrow 0.668$ (+37%)
Δ paper	-0.402

Finding: Random split improved Benign precision confirming distribution gap was a real issue. However paper specifies temporal ordering. This was exploratory — reverted in favour of within-class temporal (V10).

Version V8 — $\text{lr}=0.0001$ — CADE's learning rate tested

Files: phase2_v7.py + phase3_v8.py

Phase 2: Random stratified split (same as V7).

Phase 3: LR changed from 0.001 to 0.0001 (CADE's value, 10× smaller).

Metric	Value
Average F1	0.478
DoS-Hulk F1	0.456 — worst DoS-Hulk result
Δ paper	−0.492

Finding: Smaller lr means less convergence within 300-epoch budget. Paper's $\text{lr}=0.001$ is confirmed correct. CADE's $\text{lr}=0.0001$ is calibrated for CADE's different pair-sampling implementation.

Version V10 ★ — BEST DETECTION RESULT — 0/100 hold-out + regular L2 + within-class temporal

Files: phase2_v10.py + phase3_v10.py

Phase 2: Within-class temporal 80/20: each class sorted independently by timestamp. First 80% → train, last 20% → test. Adds Timestamp_ms as 83rd feature. Uses 0/100 open-set protocol in Phase 3 (not in Phase 2).

Phase 3: Three key changes: (1) 0/100 hold-out — zero drift rows in training. (2) Regular L2 distance $d = \sqrt{d^2 + \epsilon}$ — correct per paper Eq. 2. (3) Mean-based loss — prevents NaN from large gradient sums. similar_ratio=0.25 (CADE's value).

Metric	Value
Benign F1	0.891
Bot F1	0.878
DoS-Hulk F1	0.960
Infiltration F1	0.641
Average F1	0.842
Δ paper	−0.128
Infiltration d_final	0.020
Infiltration crossing	40.5%
Benign crossing	~98%

Finding: The 0/100 hold-out is the single most impactful change (+0.24 F1). Model never sees a single drift-class sample during training — true zero-day evaluation. DoS-Hulk F1=0.960 closely matches paper. Infiltration crossing=40.5% is our best explanation result. RECOMMENDED VERSION for reporting.

Version V11 — Exact paper specification — all-pairs mean, no similar_ratio, + explanation

Files: phase2_v10.py + phase34_v11.py

Phase 2: Same as V10 (within-class temporal, 83 features).

Phase 3: Paper's exact specification: all-pairs mean contrastive loss (no pair subsampling), regular L2 distance, 0/100 hold-out. Combined with Algorithm 2 explanation and T-SNE visualisation.

Metric	Value
Benign F1	0.891
Bot F1	0.872
DoS-Hulk F1	0.994
Infiltration F1	0.638
Average F1	0.849
Infiltration d_final	0.0097 (better than paper 0.0325)
Infiltration crossing	14%
Benign crossing	98%
DoS-Hulk crossing	0%

Finding: All-pairs mean over-compresses same-class clusters ($Q3 \approx 0.003$, upper ≈ 0.005). Excellent fidelity ($d_{\text{final}} = 0.010 < \text{paper's } 0.033$) but $d_{\text{final}} > \text{upper}$ → boundary not crossed. Compression-crossing contradiction diagnosed: the paper must use an undocumented positive-pair ratio that prevents over-compression.

Version V12 — similar_ratio=0.10 — reduces compression to widen IQR bounds

Files: phase2_v10.py + phase34_v12.py

Phase 2: Same as V10/V11.

Phase 3: Based on analysis: all-pairs (V11) over-compresses. V12 subsamples positive pairs to 10%. $n_{\text{pos_used}} = n_{\text{neg}}/9 \approx 2,408$ (from 10,710). Reduces same-class compression → wider IQR → more crossing for stuck scenarios.

Metric	Value
Benign F1	0.933
Bot F1	0.846
DoS-Hulk F1	1.000
Infiltration F1	0.624
Average F1	0.851
Bot crossing	98.77%
DoS-Hulk crossing	100%
Infiltration crossing	3%

Finding: DoS-Hulk F1=1.000 matches paper's claim exactly. Stuck scenarios (Bot, DoS-Hulk) benefit from wider IQR. Converged scenarios (Infiltration) remain over-compressed even with ratio=0.10 — non-monotonic crossing behaviour confirmed.

4. Root Cause Analysis

The paper achieves simultaneously: high F1 (converged contrastive loss, tight clusters) and high boundary crossing (moderate IQR width). All fourteen experiments show these are mutually exclusive under the published hyperparameters.

State	Q3	upper	d_final	Crossing	F1
Converged (our)	0.003–0.010	0.005–0.018	0.007–0.020	0–14%	0.84–0.85
Stuck (our)	0.3–1.5	0.5–2.5	0.10–1.90	40–100%	0.80–0.85
Paper (implied)	0.030–0.060	0.050–0.100	0.033	83.65%	0.97

The paper's converged model produces clusters $\sim 10\times$ wider than ours under identical published hyperparameters. This points to an undocumented positive-pair sampling ratio in the contrastive training loop — inherited from CADE's codebase but not documented in the DriftTrace manuscript.